Hall Ticket Number:

MR-21

Code: B414IT5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, Jan/Feb-2024

Discrete Mathematics

(Information Technology)

Time: 3 Hours

Answer ALL Questions

Max. Marks: 70

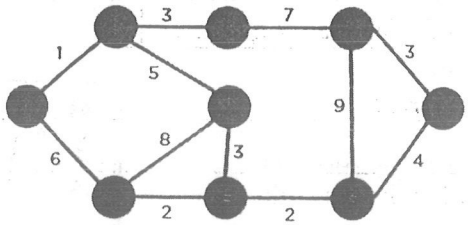
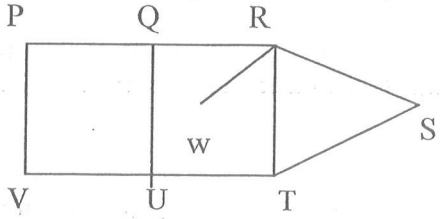
Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write the inverse, contrapositive of the following statement. P : It rains. Q: There is a flood in city.	2	1	1	1,2
1. b)	Obtain CNF of $(P \rightarrow (Q \rightarrow R)) \rightarrow (P \rightarrow (R \rightarrow Q))$.	2	3	1	1
1. c)	State principle of inclusion-Exclusion.	2	1	2	1,2
1. d)	Let X ={1,2,3,4,5,6} and R be a relation defined as (x,y) xRy $\in$ if and only if x –y is divisible by 3. Find the elements of relation of R.	2	3	2	1
1. e)	How many words of three distinct letters can be formed from CAKE?	2	2	3	1,2
1. f)	Give any three applications of Pigeonhole principle?	2	2	3	1
1. g)	A single coin is tossed 5 times. What is the probability of getting at least one head?	2	3	4	1,2
1. h)	Write the characteristic equation for the following recurrence relation $a_n - 4a_{n-4} = 0$, $n > 4$ and solve it.	2	3	4	1
1. i)	What do you mean by isomorphism? Give examples of isomorphic graphs.	2	1	5	1
1. j)	What is a planar graph? Give examples of planar and non-planar graphs.	2	2	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Obtain principal conjunctive normal form (PCNF) for the formula $(\sim p \rightarrow r) \wedge (q \leftrightarrow p)$.	5	3	1	1
2. b)	Verify the validity of the following arguments. "Every living thing is a plant or an animal. Logu"s dog is alive and it is not a plant. All animals have heart. Therefore Logu"s dog has a heart."	5	3	1	1,2
	OR				
2. c)	Using indirect proof , derive $P \rightarrow \sim S$ from $P \rightarrow Q \vee R$, $Q \rightarrow \sim P$, $S \rightarrow \sim R$, P.	5	3	1	1
2. d)	Show that $(\neg Q \wedge (P \rightarrow Q)) \rightarrow \neg P$ is a tautology.	5	2	1	1
3. a)	There are 40 computer programmers for a job. 25 know Java, 28 know Oracle and 7 know neither language. Using principle of inclusion exclusion find how many know both languages.	5	4	2	1
3. b)	Let A={a,b,c} be a set and Relation R on A is as R={(a,a),(a,b),(b,c),(c,c)}. Is R i) Reflexive ii) Symmetric iii) Transitive.	5	1	2	1
	OR				
3. c)	$f : R \rightarrow R$ is defined by $f(x) = ax + b$, for $a, b \in R$ and $a \neq 0$. Show that f is invertible and find the inverse of f.	5	2	2	1
3. d)	Draw the Hasse diagram for the partial ordering {(A,B):A$\leq$B} on the power set e(S) where S={a,b,c} and $\leq$ is subset relation.	5	2	2	1,2
4. a)	By using mathematical induction prove that the given equation is true for all positive integers. $2 + 4 + 6 + \ldots + 2n = n(n+1)$.	5	2	3	1,2
4. b)	Find the mid term of $(2x - 1/3x)^{10}$.	5	2	3	1,2

P.T.O.

	OR				
4. c)	Out of 5 consonants and 3 vowels, how many words of 3 consonants and 2 vowels can be formed?	5	1	3	1,2
4. d)	Find the number of permutations that can be made out of the letter I in word ASSASSINATION.	5	2	3	1
5. a)	Two friends Harish and Kalyan appeared for an exam. Let A be the event that Harish is selected and B is the event that Kalyan is selected. The probability of A is 2/5 and that of B is 3/7. Find the probability that both of them are selected.	5	4	4	1,2
5. b)	Find the solution for $a_n - 3a_{n-1} - 4a_{n-2} = 0$ for $n \geq 2$ and, $a_0 = a_1 = 1$.	5	2	4	1,2
	OR				
5. c)	Solve the recurrence relation $a_n=8a_{n-1}-16a_{n-2}$ for $n\geq 2$ $a_0=16$, $a_1=80$?	5	3	4	1
5. d)	A car manufacturing factory has two plants, X and Y. Plant X manufactures 70% of cars and plant Y manufactures 30%. 80% of the cars at plant X and 90% of the cars at plant Y are rated of standard quality. A car is chosen at random and is found to be of standard quality. What is the probability that it has come from plant X?	5	2	4	1
6. a)	Show that the complete bi-partite graph $K_{3,3}$ is not planar graph.	5	2	5	1,2
6. b)	What is a Hamiltonian Cycle? Draw bipartite graph $K_{3,4}$ and prove that this graph does not have a Hamiltonian cycle.	5	3	5	1
	OR				
6. c)	Define spanning tree. Apply Prim's algorithm to find minimum spanning tree on the following weighted graph.	5	3	5	1
	 Graph G(V, E)				
6. d)	Show that the following graph is verified by Euler's formula or not?	5	2	5	1
					

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome